

For this analysis, we're going to try to get means and standard errors for each time point by group for some longitudinal data. We'll use multiple imputation to do this task.

*Example - Schizophrenia

Data analyzed by Hedeker and Gibbons (1997). A randomized trial for schizophrenia. The trial contained 312 patients received drug therapy and 101 received placebo. The measurements were taken at weeks 0, 1, 3, 6, but some subjects have missing data due to dropout. The outcome of interest is severity of illness (1 = normal, ... , 7 = extremely ill);

```
data BIOS755.schizo;
input ID Group week severity;
sqrtweek = week**(0.5);
r_severity = round(severity);
datalines;
1103 1 0 5.5
1103 1 1 3.0
1103 1 3 2.5
1103 1 6 4.0
1104 1 0 6.0
1104 1 1 3.0
1104 1 3 1.5
1104 1 6 2.5
.....
9316 0 1 6.0
9316 0 3 6.5
9316 0 6 6.0
run;
```

* To run the imputation, we first want to change the data from long to wide:
* Going from long to wide (this will only output the outcome and id):

```
proc transpose data=BIOS755.schizo out=schizo_wide prefix=Y;
by id;
id week;
var severity;
run;
```

* Merging the wide outcome data with the other covariate data:

```
data BIOS755.schizo_wide;
merge BIOS755.schizo schizo_wide;
by id;
if first.id;
drop week severity sqrtweek r_severity _NAME_ _LABEL_;
run;
```

```
proc print data = BIOS755.schizo_wide (obs=10);
run;
```

Obs	ID	Group	Y0	Y1	Y3	Y6
1	1103	1	5.5	3.0	2.5	4.0
2	1104	1	6.0	3.0	1.5	2.5
3	1105	1	4.0	3.0	1.0	.

Obs	ID	Group	Y0	Y1	Y3	Y6
4	1106	1	3.0	1.0	1.5	1.0
5	1107	0	5.0	5.0	5.0	5.5
6	1108	1	6.0	6.0	3.5	4.5
7	1109	1	4.0	2.0	2.0	2.5
8	1110	1	4.0	4.5	4.0	3.5
9	1111	1	5.5	5.5	5.5	2.5
10	1113	1	4.0	2.5	4.5	3.0

```
proc sort data = BIOS755.schizo_wide;
by group;
run;
```

```
* Running the imputation on the wide data;
* First have an imputation model for Y1 (Y0 has no missingness);
* Then we have an imputation model for Y3 given Y0, group, and the imputed
Y1;
* Then we have an imputation model for Y6 given Y0, group, and the imputed Y1
& Y3;
```

```
proc mi data=BIOS755.schizo_wide NIMPUTE=10 seed=8675309 out=MI_wide;
class group;
fcs nbiter=20 plots=trace regpmm (Y1 = Y0 group Y0*group/ k = 8);
fcs nbiter=20 plots=trace regpmm (Y3 = Y1 Y0 group Y0*group Y1*group/ k = 8);
fcs nbiter=20 plots=trace regpmm (Y6 = Y3 Y1 Y0 group Y0*group Y1*group
Y3*group/ k = 8);
var Y0 group Y1 Y3 Y6;
run;
```

Imputation Methods

[DISCRIM](#)

Specifies the discriminant function method

[LOGISTIC](#)

Specifies the logistic regression method

[REG](#)

Specifies the regression method

[REGPMM](#)

Specifies the predictive mean matching method

ORDER=FREQ | VAR

specifies the variable ordering in which to impute missing values in the filled-in and imputation phases. The ORDER=FREQ option orders the variables by the descending frequency counts of variables and the ORDER=VAR orders the variables as specified in the VAR statement. The default is ORDER=FREQ.

The MI Procedure

Model Information

Data Set	BIOS755.SCHIZO_WIDE
Method	FCS
Number of Imputations	10
Number of Burn-in Iterations	20
Seed for random number generator	8675309

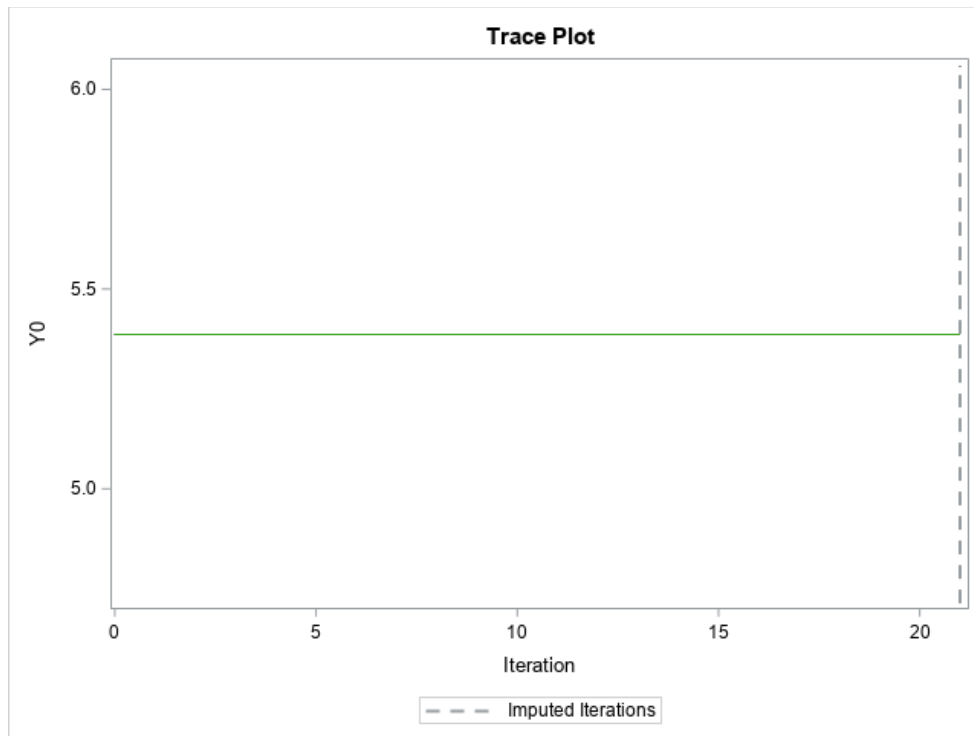
FCS Model Specification

Method	Imputed Variables
Regression	Y0
Regression-PMM(K= 8)	Y1 Y3 Y6
Discriminant Function	Group

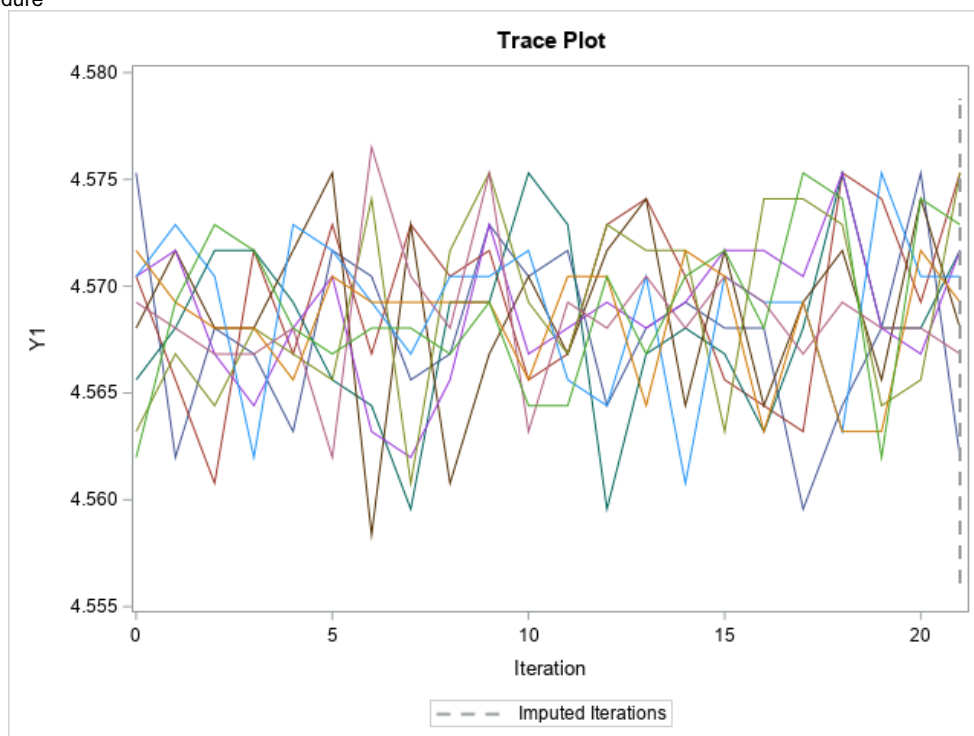
Missing Data Patterns

Group	Y0	Group	Y1	Y3	Y6	Freq	Percent	Group Means			
								Y0	Y1	Y3	Y6
1	X	X	X	X	X	312	75.54	5.379808	4.562821	4.092949	3.349038
2	X	X	X	X	.	53	12.83	5.320755	4.500000	3.641509	.
3	X	X	X	.	.	45	10.90	5.566667	4.744444	.	.
4	X	X	.	.	.	3	0.73	4.500000	.	.	.

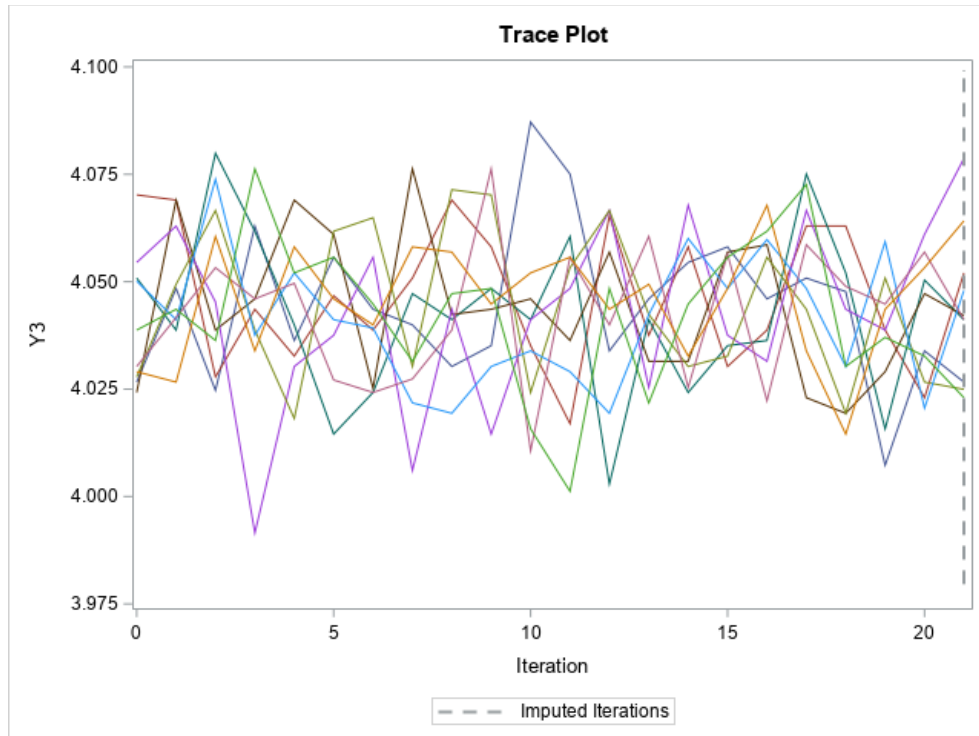
The MI Procedure



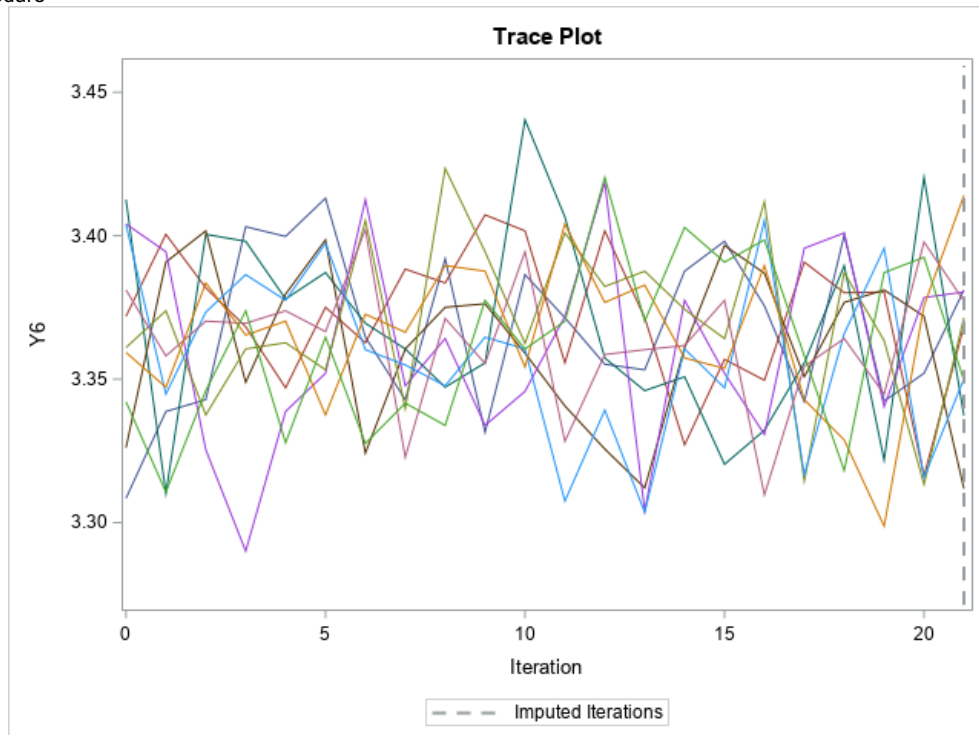
The MI Procedure



The MI Procedure



The MI Procedure



Variance Information (10 Imputations)								
Variable	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency	
	Between	Within	Total					
Y1	0.000016432	0.003695	0.003714	407.58	0.004891	0.004873	0.999513	
Y3	0.000313	0.005299	0.005643	332.17	0.064940	0.061755	0.993862	
Y6	0.000794	0.005859	0.006733	214	0.149081	0.132977	0.986877	

Parameter Estimates (10 Imputations)										
Variable	Mean	Std Error	95% Confidence Limits		DF	Minimum	Maximum	Mu0	t for H0: Mean=Mu0	Pr > t
Y1	4.570339	0.060939	4.450545	4.690133	407.58	4.561985	4.575303	0	75.00	<.0001
Y3	4.044189	0.075123	3.896412	4.191965	332.17	4.023002	4.078692	0	53.83	<.0001
Y6	3.364165	0.082054	3.202427	3.525903	214	3.311864	3.414044	0	41.00	<.0001

```
proc print data = MI_wide (obs=10);
run;
```

Obs	_Imputation_	ID	Group	Y0	Y1	Y3	Y6
1	1	1107	0	5.0	5.0	5.0	5.5
2	1	1115	0	5.5	4.5	5.5	4.5
3	1	1124	0	5.0	5.0	4.0	4.5
4	1	1129	0	6.0	4.5	3.5	2.0
5	1	1136	0	5.0	5.5	4.5	5.5
6	1	1140	0	5.5	6.0	6.0	6.5
7	1	1301	0	5.0	2.5	2.5	3.5
8	1	1306	0	4.5	5.0	1.5	2.5
9	1	1309	0	6.0	6.0	3.5	2.0
10	1	1313	0	6.5	6.0	6.0	6.0

* Now we'll do a simple analysis (means and standard errors) for each timepoint based on the imputed data;

```
* First, we'll analyze the data by _imputation_;
proc means data=MI_wide mean stderr;
by _imputation_ group;
var Y0 Y1 Y3 Y6;
output out=MI_mn mean=mn_Y0 mn_Y1 mn_Y3 mn_Y6 stderr=SE_Y0 SE_Y1 SE_Y3 SE_Y6;
run ;
```

```
proc sort data=MI_mn ;
by group _imputation_ ;
```

```

run ;

* Second, we'll use mianalyze to summarize;
proc mianalyze data=MI_mn ;
by group ;
modeleffects mn_Y0 mn_Y1 mn_Y3 mn_Y6 ;
stderr SE_Y0 SE_Y1 SE_Y3 SE_Y6;
ods output parameterestimates=outcombine_1 ;
run ;

```

The SAS System

The MIANALYZE Procedure
Group=0

Model Information

Data Set WORK.MI_MN

Number of Imputations 10

Variance Information (10 Imputations)

Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency
	Between	Within	Total				
mn_Y0	0	0.006536	0.006536	.	0	.	.
mn_Y1	0	0.013068	0.013068	.	0	.	.
mn_Y3	0.000465	0.013100	0.013611	6379.7	0.039025	0.037861	0.996228
mn_Y6	0.008918	0.019539	0.029349	80.563	0.502033	0.350170	0.966168

Parameter Estimates (10 Imputations)

Parameter	Estimate	Std Error	95% Confidence Limits		DF	Minimum	Maximum	Theta0	t for H0: Parameter=Theta0	Pr > t
mn_Y0	5.373267	0.080848	.	.	.	5.373267	5.373267	0	.	.
mn_Y1	4.990099	0.114317	.	.	.	4.990099	4.990099	0	.	.
mn_Y3	4.862475	0.116667	4.633768	5.091183	6379.7	4.829703	4.897030	0	41.68	<.0001
mn_Y6	4.492574	0.171315	4.151682	4.833466	80.563	4.358416	4.625743	0	26.22	<.0001

The MIANALYZE Procedure
Group=1

Model Information

Data Set WORK.MI_MN

Number of Imputations 10

Variance Information (10 Imputations)

Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency
	Between	Within	Total				
mn_Y0	0	0.002423	0.002423	.	0	.	.
mn_Y1	0.000028793	0.004877	0.004909	216214	0.006494	0.006461	0.999354

Variance Information (10 Imputations)

Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency
	Between	Within	Total				
mn_Y3	0.000816	0.007005	0.007903	697.41	0.128158	0.116130	0.988520
mn_Y6	0.001135	0.006478	0.007727	344.47	0.192805	0.166465	0.983626

Parameter Estimates (10 Imputations)

Parameter	Estimate	Std Error	95% Confidence Limits		DF	Minimum	Maximum	Theta0	t for H0: Parameter=Theta0	Pr > t
mn_Y0	5.390385	0.049222	.	.	.	5.390385	5.390385	0	.	.
mn_Y1	4.434455	0.070064	4.297131	4.571779	216214	4.423397	4.441026	0	63.29	<.0001
mn_Y3	3.779295	0.088898	3.604755	3.953835	697.41	3.742628	3.832372	0	42.51	<.0001
mn_Y6	2.998878	0.087904	2.825981	3.171775	344.47	2.952564	3.054167	0	34.12	<.0001

```
proc print data=outcombine_1;
run;
```

Obs	Group	NImpute	Parm	Estimate	StdErr	LCLMean	UCLMean	DF	Min	Max	The ta0	tValue	Probt
1	0	10	mn_Y0	5.373267	0.080848	.	.	.	5.373267	5.373267	0	.	.
2	0	10	mn_Y1	4.990099	0.114317	.	.	.	4.990099	4.990099	0	.	.
3	0	10	mn_Y3	4.862475	0.116667	4.633768	5.091183	6379.7	4.829703	4.897030	0	41.68	<.0001
4	0	10	mn_Y6	4.492574	0.171315	4.151682	4.833466	80.563	4.358416	4.625743	0	26.22	<.0001
5	1	10	mn_Y0	5.390385	0.049222	.	.	.	5.390385	5.390385	0	.	.
6	1	10	mn_Y1	4.434455	0.070064	4.297131	4.571779	216214	4.423397	4.441026	0	63.29	<.0001
7	1	10	mn_Y3	3.779295	0.088898	3.604755	3.953835	697.41	3.742628	3.832372	0	42.51	<.0001
8	1	10	mn_Y6	2.998878	0.087904	2.825981	3.171775	344.47	2.952564	3.054167	0	34.12	<.0001

```
proc means data = BIOS755.schizo_wide mean stderr clm alpha=0.05;
var Y0 Y1 Y3 Y6;
by group;
run;
```

The MEANS Procedure				
Group=0				
Variable	Mean	Std Error	Lower 95% CL for Mean	Upper 95% CL for Mean
Y0	5.3732673	0.0808481	5.2128670	5.5336677
Y1	4.9900990	0.1143168	4.7632978	5.2169002
Y3	4.7927711	0.1234220	4.5472455	5.0382967
Y6	4.3156250	0.1781494	3.9596219	4.6716281
Group=1				
Variable	Mean	Std Error	Lower 95% CL for Mean	Upper 95% CL for Mean
Y0	5.3903846	0.0492222	5.2935341	5.4872352
Y1	4.4388350	0.0703168	4.3004728	4.5771971
Y3	3.8021277	0.0867251	3.6314143	3.9728411
Y6	3.0995968	0.0892931	2.9237238	3.2754698

```

data schizo_long;
set MI_wide;
by _imputation_;
  week=0;
  severity=Y0;
sqrtweek = week**(0.5);
r_severity = round(severity);
  output;
  week=1;
  severity=Y1;
sqrtweek = week**(0.5);
r_severity = round(severity);
  output;
  week=3;
  severity=Y3;
sqrtweek = week**(0.5);
r_severity = round(severity);
  output;
  week = 6;
  severity=Y6;
sqrtweek = week**(0.5);
r_severity = round(severity);
  output;
output;
drop Y0 - Y6;
run;

```

```

proc glimmix data=schizo_long method=QUAD(qpoints=20);
by _imputation_;
class ID;
model r_severity = Group sqrtweek Group*sqrtweek/solution
link=cumlogit dist=multinomial covb;
random intercept/subject=ID type=UN G;
ods output ParameterEstimates=mixparms CovB=mixcovb;
run;
quit;

```

The GLIMMIX Procedure
Imputation Number=1

Model Information

Data Set	WORK.SCHIZO_LONG
-----------------	------------------

Model Information		
Response Variable		r_severity
Response Distribution		Multinomial (ordered)
Link Function		Cumulative Logit
Variance Function		Default
Variance Matrix Blocked By		ID
Estimation Technique		Maximum Likelihood
Likelihood Approximation		Gauss-Hermite Quadrature
Degrees of Freedom Method		Containment
Class Level Information		
Class	Levels	Values
ID	413	1103 1104 9316
Number of Observations Read		2065
Number of Observations Used		2065
Response Profile		
Ordered Value	r_severity	Total Frequency
The GLIMMIX procedure is modeling the probabilities of levels of r_severity having lower Ordered Values in the Response Profile table.		
1	1	113
2	2	214
3	3	265
4	4	374

Response Profile					
Ordered Value	r_severity	Total Frequency			
5	5	473			
6	6	523			
7	7	103			
Dimensions					
G-side Cov. Parameters	1				
Columns in X	9				
Columns in Z per Subject	1				
Subjects (Blocks in V)	413				
Max Obs per Subject	5				
Optimization Information					
Optimization Technique		Dual Quasi-Newton			
Parameters in Optimization		10			
Lower Boundaries		1			
Upper Boundaries		0			
Fixed Effects		Not Profiled			
Starting From		GLM estimates			
Quadrature Points		20			
Iteration History					
Iteration	Restarts	Evaluations	Objective Function	Change	Max Gradient
0	0	4	6246.2209238	.	181.2061

Iteration History					
Iteration	Restarts	Evaluations	Objective Function	Change	Max Gradient
1	0	2	6119.7986792	126.42224456	148.073
2	0	2	6084.2757253	35.52295396	358.4737
3	0	2	6039.7454613	44.53026400	209.4177
4	0	3	6021.6823831	18.06307819	122.7434
5	0	2	5989.8411125	31.84127061	89.32346
6	0	2	5944.9999945	44.84111795	81.78881
7	0	3	5925.9813321	19.01866246	63.43059
8	0	3	5916.7433289	9.23800313	23.42529
9	0	3	5911.3717035	5.37162538	15.65413
10	0	3	5910.1900331	1.18167041	20.30721
11	0	4	5906.7392412	3.45079196	18.64359
12	0	3	5905.1062513	1.63298991	6.520598
13	0	3	5904.707514	0.39873724	5.402078
14	0	3	5904.6197756	0.08773845	1.510322
15	0	3	5904.6130912	0.00668441	0.25839
16	0	3	5904.6129354	0.00015580	0.036091
17	0	3	5904.6129341	0.00000130	0.00516
Convergence criterion (GCONV=1E-8) satisfied.					
Fit Statistics					
-2 Log Likelihood		5904.61			

Fit Statistics	
AIC (smaller is better)	5924.61
AICC (smaller is better)	5924.72
BIC (smaller is better)	5964.85
CAIC (smaller is better)	5974.85
HQIC (smaller is better)	5940.53

Fit Statistics for Conditional Distribution	
-2 log L(r_severity r. effects)	4734.78

Estimated G Matrix		
Effect	Row	Col1
Intercept	1	4.3985

Covariance Parameter Estimates			
Cov Parm	Subject	Estimate	Standard Error
UN(1,1)	ID	4.3985	0.4306

Solutions for Fixed Effects						
Effect	r_severity	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	1	-7.8791	0.3365	411	-23.41	<.0001
Intercept	2	-5.9237	0.3098	411	-19.12	<.0001
Intercept	3	-4.4812	0.2967	411	-15.11	<.0001
Intercept	4	-2.8171	0.2843	411	-9.91	<.0001
Intercept	5	-0.7281	0.2756	411	-2.64	0.0086

Solutions for Fixed Effects						
Effect	r_severity	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	6	2.9675	0.2935	411	10.11	<.0001
Group		0.04435	0.3118	1645	0.14	0.8869
sqrtweek		0.6120	0.09639	1645	6.35	<.0001
Group*sqrtweek		1.3676	0.1135	1645	12.05	<.0001

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Group	1	1645	0.02	0.8869
sqrtweek	1	1645	40.32	<.0001
Group*sqrtweek	1	1645	145.24	<.0001

```
proc print data = mixparms;
run;
```

Obs	_Imputation_	Effect	r_severity	Estimate	StdErr	DF	tValue	Probt
1	1	Intercept	1	-7.8791	0.3365	411	-23.41	<.0001
2	1	Intercept	2	-5.9237	0.3098	411	-19.12	<.0001
3	1	Intercept	3	-4.4812	0.2967	411	-15.11	<.0001
4	1	Intercept	4	-2.8171	0.2843	411	-9.91	<.0001
5	1	Intercept	5	-0.7281	0.2756	411	-2.64	0.0086
6	1	Intercept	6	2.9675	0.2935	411	10.11	<.0001
7	1	Group	—	0.04435	0.3118	1645	0.14	0.8869
8	1	sqrtweek	—	0.6120	0.09639	1645	6.35	<.0001

Obs	_Imputation_	Effect	r_severity	Estimate	StdErr	DF	tValue	Probt
9	1	Group*sqrtweek	—	1.3676	0.1135	1645	12.05	<.0001
10	2	Intercept	1	-7.8142	0.3329	411	-23.47	<.0001
11	2	Intercept	2	-5.7813	0.3061	411	-18.89	<.0001
12	2	Intercept	3	-4.3582	0.2934	411	-14.85	<.0001
13	2	Intercept	4	-2.7541	0.2816	411	-9.78	<.0001
14	2	Intercept	5	-0.7094	0.2733	411	-2.60	0.0098
15	2	Intercept	6	2.9659	0.2926	411	10.14	<.0001

```

proc mianalyze parms=mixparms covb(effectvar=rowcol)=mixcovb
;
  *The covb is only needed for multi-variate results (i.e.,
  type III tests or contrast statements;
  modeleffects Group sqrtweek Group*sqrtweek;
  title 'MIANALYZE Results';
run;

```

MIANALYZE Results

The MIANALYZE Procedure

Model Information							
PARMS Data Set	WORK.MIXPARMS						
COVB Data Set	WORK.MIXCOVB						
Number of Imputations	10						
Variance Information (10 Imputations)							
Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency
	Between	Within	Total				
intercept	0.002157	0.085704	0.088076	12401	0.027686	0.027097	0.997298

Variance Information (10 Imputations)									
Parameter	Variance			DF	Relative Increase in Variance	Fraction Missing Information	Relative Efficiency		
	Between	Within	Total						
Group	0.000994	0.095855	0.096948	70810	0.011402	0.011302	0.998871		
sqrtweek	0.002798	0.009201	0.012278	143.28	0.334456	0.260877	0.974576		
Group*sqrtweek	0.003340	0.012673	0.016346	178.2	0.289877	0.233289	0.977203		

Parameter Estimates (10 Imputations)									
Parameter	Estimate	Std Error	95% Confidence Limits		DF	Minimum	Maximum	Theta0	t for H0: Parameter=Theta0 Pr > t
intercept	2.935096	0.296777	2.35337	3.516825	12401	2.847742	2.999106	0	9.89 <.0001
Group	0.060386	0.311365	-0.54989	0.670660	70810	0.001450	0.118389	0	0.19 0.8462
sqrtweek	0.692863	0.110807	0.47383	0.911892	143.28	0.602466	0.757790	0	6.25 <.0001
Group*sqrtweek	1.304536	0.127853	1.05224	1.556837	178.2	1.208516	1.376741	0	10.20 <.0001

From non-imputed data:

Solutions for Fixed Effects							
Effect	r_severity	Group	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	1		-5.3218	0.2523	411	-21.09	<.0001
Intercept	2		-3.8528	0.2188	411	-17.61	<.0001
Intercept	3		-2.8738	0.2090	411	-13.75	<.0001
Intercept	4		-1.7987	0.2015	411	-8.93	<.0001
Intercept	5		-0.4393	0.1951	411	-2.25	0.0249
Intercept	6		1.9550	0.2127	411	9.19	<.0001
Group		1	-0.01599	0.2220	1080	-0.07	0.9426
Group		0	0	.	.	.	.
sqrtweek			0.4862	0.1138	1080	4.27	<.0001

Solutions for Fixed Effects							
Effect	r_severity	Group	Estimate	Standard Error	DF	t Value	Pr > t
sqrtweek*Group		1	0.8172	0.1299	1080	6.29	<.0001
sqrtweek*Group		0	0	.	.	.	.

Type III Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
Group	1	1080	0.01	0.9426
sqrtweek	1	1080	176.53	<.0001
sqrtweek*Group	1	1080	39.60	<.0001

Why are they different?

GLMM provides unbiased estimates under MAR **given the model is correctly specified**. However, if dropout depends on things not in our model, then the observed data are biased toward certain types of subjects, and GLMM and imputation will differ because they make different assumptions about the missing outcomes.

For example, patients who are not improving (i.e., still have high severity scores) are more likely to drop out, especially in the placebo group. But our model only has group, sqrtweek, and group*sqrtweek. This can result in the placebo group looks like it is improving more than it truly is→ The treatment effect (group×time) is **biased toward no effect**.

The imputation model, however, is different. It doesn't just have group, sqrtweek, and group*sqrtweek. So, it give different (probably better) estimates.